Name: ..... **KAUST ID:** .....

Assessment Form for HW3

Functionality	Item	Subtasks	Points	Note
Data Parsing	`DP-1`	Classification dataset (all synthetic data, Iris and Titanic)	3	
	`DP-2`	Regression dataset (all synthetic data, and at least two datasets among Automobile, Forest Fires, and Wine)	3	
Forward Propagation	`FP-1`	Neural structure (one hidden layer and one output layer)	5	
	`FP-2`	Three activation functions (Sigmoid, Hyperbolic Tangent, ReLU)	9	
Backpropagation (support all activations)	`BP-1`	MSE cost and cross-entropy loss	6	
	`BP-2`	Gradient computation	12	
	`BP-3`	Weight updates	10	
Experiments & Reporting *(Regression)*	`ER-1`	Plot MSE over iterations for the training dataset; output MSE for the test dataset	6	
	`ER-2`	Plot prediction vs. ground truth over data samples	2	
	`ER-3`	Experiment with different numbers of hidden units, learning rates, and activation functions; report results in a table or figure	6	
	`ER-4`	Compare performance (MSE, R^2, training speed) across settings and **conclude which configuration is best**	6	
Experiments & Reporting *(Classification)*	`EC-1`	Plot cross-entropy loss over iterations for training; output cross-entropy loss on the test dataset	3	
	`EC-2`	Plot the confusion matrix for each dataset	3	
	`EC-3`	Experiment with different hidden units, learning rates, and activation functions; report results in a table or figure	6	
	`EC-4`	**Benchmark accuracy comparison** (see Table 1): full marks if within ± 0.10 of benchmark; partial credit scaled linearly, reaching 0 at -0.20 gap	20	

Table 1: Classification Benchmark Accuracies (reference implementation)

Dataset	Benchmark Acc.	Full-credit threshold	Scoring rule
Iris (binary)	1.000	≥ 0.900	Linear: 4 pts $\times \max\left(0,\ 1 - \frac{\Delta}{0.20}\right)$
Titanic	0.832	≥ 0.732	Linear: 4 pts $\times \max\left(0,\ 1 - \frac{\Delta}{0.20}\right)$
Synth Linear 2D	0.950	≥ 0.850	Linear: 4 pts $\times \max\left(0,\ 1 - \frac{\Delta}{0.20}\right)$
Synth XOR 2D	0.960	≥ 0.860	Linear: 4 pts $\times \max\left(0,\ 1 - \frac{\Delta}{0.20}\right)$
Synth Circles 2D	1.000	≥ 0.900	Linear: 4 pts $\times \max\left(0,\ 1 - \frac{\Delta}{0.20}\right)$

$\Delta = \max(0,\ \text{benchmark} - \text{student accuracy})$; within ± 0.10 of benchmark receives full marks; score reaches 0 at gap $= 0.20$.

Notes:

1. You may develop functionality beyond the requirements (e.g., parallelism via `std::execution::par`, additional neural network layers). Extra credit is awarded based on functional complexity and demonstrated understanding. **Examples: parallelism, more neural layers.**
2. Extra points can compensate for lost points on compulsory items across all assignments.
3. You can earn a maximum of **20 extra points** per assignment.